

PROBABILITY THEORY

MTM2592-Gr:2

FOURTH WEEK

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Ex: If $F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$, find $f(x)$.

Solution:

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases} \quad \text{For } 0 < x < 1 \quad \frac{dF}{dx} = f(x) \Rightarrow \frac{d(x)}{dx} = 1 = f(x)$$

Example: Probability function of discrete random variable of X is given below;

$X=x:$	-3	0	2	3
$P(X=x):$	0,2	0,1	0,4	c

- Find the value of c?
- At which value, X's probability maximum?
- $P(X>0)=?$
- $P(X=-2)=?$
- Find the cumulative function of X?
- Show F(x) as a Table.
- Draw the Graphs of f(x) and F(x)

Solution:

a) $0,2+0,1+0,4+c=1 \implies c=0,3$

b) 2

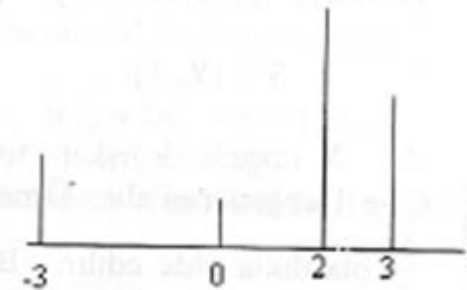
c) $P(X>0) = P(X=2)+P(X=3)=0,4+0,3=0,7.$

d) $P(X = -2) = 0.$

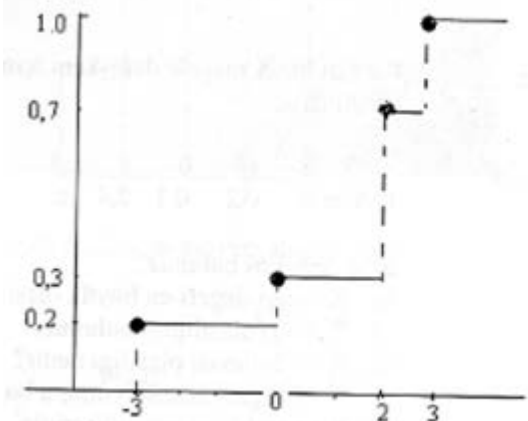
e)
$$F(x) = \begin{cases} 0 & x < -3 \\ 0,2 & -3 \leq x < 0 \\ 0,3 & 0 \leq x < 2 \\ 0,7 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

f)

$X=x:$	-3	0	2	3
$F(x)=P(X \leq x):$	0,2	0,3	0,7	1.0



Graphs of f(x)



Graphs of cumulative function of F(x)

Example: A coin is tossed until a head occurs. Let X be a random variable that shows necessary number of tossing. Find the probability function of random variable X.

Solution:

		Probability
First toss	H	1/2
Second toss	TH	1/2.1/2
Third toss	TTH	1/2.1/2.1/2
.	.	.
N-th toss	TTT...TH	$(1/2)^{N-1}.1/2$
.	.	.

X = x	1	2	...	x	...
$f(x) = P(X=x)$	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	...	$\left(\frac{1}{2}\right)^x$	...

$$f(x) = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, \dots$$

Note: $a + ar + ar^2 + \dots = \frac{a}{1-r}$

$$(-1 \leq r \leq 1)$$

$$\sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1}{2} + \frac{1}{2^2} + \dots$$

$$a = \frac{1}{2}, r = \frac{1}{2} \Rightarrow \sum_{x=1}^{\infty} f(x) = \frac{1/2}{1-1/2} = 1$$

EXPECTED VALUE:

Definition: Let X be a random variable with probability function $f(x)$. The expected value (or mean) of X is

$$E(X) = \mu_X = \sum_{i=1}^N x_i f(x_i)$$



if X is discrete

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x \cdot f(x) dx$$



if X is continuous.

Theorem: If $g(x)$ is a function of random variable X with probability function $f(x)$, then expected value of $g(x)$ is

$$E(g(x)) = \mu_{g(x)} = \begin{cases} \sum_{i=1}^n g(x_i) f(x_i), & \text{for discrete variable} \\ \int_{-\infty}^{\infty} g(x) f(x) dx, & \text{for continuous variable} \end{cases}$$

Properties of Expected Value

c is a constant, X and Y are random variables;

1) $E(c)=c$

2) $E(X+c)=E(X)+c$

3) $E(cX)=cE(X)$

4) $x \geq 0 \Rightarrow E(X) \geq 0$

5) $x \leq y \Rightarrow E(X) \leq E(Y)$

6) $E(X+Y) = E(X) + E(Y)$

7) $|E(X)| \leq E(|X|)$

8) Under the assumption that X and Y are independent

$$E(XY) = E(X) E(Y)$$

9) $E(X^2) \neq [E(X)]^2$

10) The expected value E is a linear operator, where a and b constant.

$$E(aX+b) = a E(X) + b$$

Example: Prove that the expected value E is a linear operator, where a and b constant.

$$E(aX + b) = a E(X) + b$$

Solution:

$$E(aX + b) = \sum (aX + b)f(x) = a \sum Xf(x) + b \sum f(x) = aE(x) + b$$

Example: Show $E(cX) = cE(X)$

Solution:

$$\begin{aligned} E(cX) &= \sum_{i=1}^k cx_i P(X = x_i) \\ &= c \sum_{i=1}^k x_i P(X = x_i) \\ &= cE(x) \end{aligned}$$

Example:

X	0	1	2
f(x)	1/2	1/4	1/4

a) $E(X)=?$

b) If $Y=g(X)=2x^2+x$, then find $E(Y)=?$

Solution:

Example: In a game, a coin is successively tossed until it shows a tail or head 3 times. If the game is over with 3 or more trials, then the player will win 15 tl, otherwise the player will lose 5 tl. Find the expected value of X that denotes the gain or loss of the player?

Solution:

Example:

According to a game, a die is rolled and if it shows a prime number, the player wins two times of the number. Otherwise, the player loses 1.9 times of the showing number. Would the player play this game?

Solution:

EXAMPLE

If X is the number of points rolled with a balanced die, find the expected value of $g(X) = 2X^2 + 1$.

Solution

Since each possible outcome has the probability $\frac{1}{6}$, we get

$$\begin{aligned} E[g(X)] &= \sum_{x=1}^6 (2x^2 + 1) \cdot \frac{1}{6} \\ &= (2 \cdot 1^2 + 1) \cdot \frac{1}{6} + \cdots + (2 \cdot 6^2 + 1) \cdot \frac{1}{6} \\ &= \frac{94}{3} \end{aligned}$$

EXAMPLE

If X has the probability density

$$f(x) = \begin{cases} e^x & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X) = e^{3X/4}$.

Solution:

$$\begin{aligned} E[e^{3X/4}] &= \int_0^{\infty} e^{3x/4} \cdot e^{-x} dx \\ &= \int_0^{\infty} e^{-x/4} dx \\ &= 4 \end{aligned}$$

EXAMPLE

If the probability density of X is given by

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

(a) show that

$$E(X^r) = \frac{2}{(r+1)(r+2)}$$

(b) and use this result to evaluate

$$E[(2X+1)^2]$$

Solution

(a)

$$\begin{aligned} E(X^r) &= \int_0^1 x^r \cdot 2(1-x) dx = 2 \int_0^1 (x^r - x^{r+1}) dx \\ &= 2 \left(\frac{1}{r+1} - \frac{1}{r+2} \right) = \frac{2}{(r+1)(r+2)} \end{aligned}$$

(b) Since $E[(2X+1)^2] = 4E(X^2) + 4E(X) + 1$ and substitution of $r = 1$ and $r = 2$ into the preceding formula yields $E(X) = \frac{2}{2 \cdot 3} = \frac{1}{3}$ and $E(X^2) = \frac{2}{3 \cdot 4} = \frac{1}{6}$, we get

$$E[(2X+1)^2] = 4 \cdot \frac{1}{6} + 4 \cdot \frac{1}{3} + 1 = 3$$

EXAMPLE

Show that

$$E[(aX + b)^n] = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i E(X^{n-i})$$

Solution

Since $(ax + b)^n = \sum_{i=0}^n \binom{n}{i} (ax)^{n-i} b^i$, it follows that

$$\begin{aligned} E[(aX + b)^n] &= E \left[\sum_{i=0}^n \binom{n}{i} a^{n-i} b^i X^{n-i} \right] \\ &= \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i E(X^{n-i}) \end{aligned}$$

VARIANCE AND STANDARD DEVIATION:

Definition: If X is a random variable with probability function $f(x)$ and expected value μ , then the variance of X is

$$V(X) = \text{Var}(X) = \sigma_X^2 = E[(X - \mu)^2] = \begin{cases} \sum_x (X - \mu)^2 f(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (X - \mu)^2 f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Theorem: The variance of a random variable X is

$$\text{Var}(X) = E(X^2) - \mu^2 \quad (\mu^2 = E^2(X))$$

Proof: $\text{Var}(X) = \sigma_X^2 = E[(X - \mu)^2] = \sum [x - E(X)]^2 f(x)$

$$\sum_x [x - E(X)]^2 f(x) = \sum_x x^2 f(x) - 2\mu \sum_x x \cdot f(x) + \mu^2 \sum_x f(x)$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$= E(X^2) - \mu^2$$

$$\text{Var}(X) = \sigma_X^2 = E(X^2) - [E(X)]^2$$

Properties of Variance;

If c is a constant, X is a random variable,

1) $\text{Var}(c)=0$

Proof:
$$\text{Var}(c) = \frac{1}{k} \sum_{i=1}^k (c)^2 - \left[\frac{1}{k} \sum_{i=1}^k (c) \right]^2 = \frac{1}{k} \cdot k \cdot c^2 - \left(\frac{1}{k} \cdot k \cdot c \right)^2 = c^2 - c^2 = 0$$

2) $\text{Var}(X+c)=\text{Var}(X)$

Proof:
$$\begin{aligned} \text{Var}(X+c) &= \frac{1}{k} \sum_{i=1}^k (x_i + c)^2 - \left[\frac{1}{k} \sum_{i=1}^k (x_i + c) \right]^2 = \frac{1}{k} \sum_{i=1}^k (x_i^2 + 2cx_i + c^2) - (\bar{x} + c)^2 \\ &= \frac{1}{k} \sum_{i=1}^k x_i^2 + \frac{2c}{k} \sum_{i=1}^k x_i + \frac{1}{k} \cdot k \cdot c^2 - \bar{x}^2 - c^2 - 2c\bar{x} \\ &= \frac{1}{k} \sum_{i=1}^k x_i^2 - \bar{x}^2 = \text{Var}(X) \end{aligned}$$

3) $\text{Var}(cX)=c^2\text{Var}(X)$

Proof:
$$\begin{aligned} \text{Var}(cX) &= \frac{1}{k} \sum_{i=1}^k (cx_i)^2 - \left[\frac{1}{k} \sum_{i=1}^k (cx_i) \right]^2 = \frac{c^2}{k} \sum_{i=1}^k x_i^2 - (c\bar{x})^2 \\ &= \frac{c^2}{k} \sum_{i=1}^k x_i^2 - c^2 \bar{x}^2 = c^2 \text{Var}(X) \end{aligned}$$

4) $\text{Var}(aX+b)=a^2\text{Var}(X)$

Proof: Let, $Y=aX+b$, then

$$\begin{aligned}\text{Var}(Y) &= E\left[\left(Y - E(Y)\right)^2\right] = E\left[\left(aX + b - E(aX + b)\right)^2\right] \\ &= E\left[\left(aX + b - aE(X) - b\right)^2\right] \\ &= E\left[a^2(X - E(X))^2\right] = a^2E\left[(X - \mu)^2\right] = a^2\text{Var}(X)\end{aligned}$$

Var is not a linear operator!

Definition: The positive square root of the variance is called the **standard deviation of X**.

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Example:

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & \text{if } 0 < x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

- a) Show that f is a probability density function.
- b) Determine the cumulative distribution $F(x)$
- c) Find the probabilities;

$$P(X \leq 3/2), \quad P(X > 1/2), \quad P(0 < X \leq 7/4) \text{ and } P(1/2 < X \leq 9/2)$$

Solution:

Example:

$$f(x) = \begin{cases} k(2x + 4), & \text{if } 0 < x \leq 6 \\ 0, & \text{elsewhere} \end{cases}$$

- a) If f is a probability density function, find the value of k .
- b) Compute the mean $E(x)=\mu$ and standard deviation σ .
- c) Determine the cumulative function $F(x)$.
- d) Find the probabilities;

$$P(3 < X \leq 5), \quad P(X \geq 9/2), \quad P(|X - 5| \leq 3), \quad P(X \leq 3 | 1 \leq X \leq 4) = ?$$

Solution:

Example:

$$f(x) = \begin{cases} \frac{3}{8}x, & \text{if } 0 < x \leq 2 \\ \frac{1}{12}, & \text{if } 2 < x \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

- a) Show that $f(x)$ is a probability density function.
- b) Find the cumulative distribution $F(x)$
- c) Compute

$$P(1/4 \leq X \leq 5/2), \quad \text{and} \quad P(|X| \geq 1)$$

Solution:

MOMENTS

Definition: The r^{th} moment about the origin of a random variable X , denoted by m_r , is the expected value of x^r ; symbolically

$$m_r = E[(X)^r] = \begin{cases} \sum_{i=1}^n (x_i)^r \cdot f(x_i) & \text{If } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x)^r \cdot f(x) dx & \text{If } X \text{ is continuous} \end{cases}$$

where $r=0,1,2,\dots$

Definition: The expected value of $(X-a)^r$, namely $E[(X-a)^r]$ is called the r^{th} moment about a of X .

$$\mu_r = E[(X-a)^r] = \begin{cases} \sum_{i=1}^n (x_i - a)^r \cdot f(x_i) & \text{If } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x-a)^r \cdot f(x) dx & \text{If } X \text{ is continuous} \end{cases}$$

In order to exist the moments, the sum and the integral in the formula must be defined.

$$r = 0 \Rightarrow E(X^0) = 1 = m_0$$

$$r = 1 \Rightarrow E(X) = \mu = m_1$$

$$r \Rightarrow, E(X^r) = m_r$$

$$\begin{aligned}\mu_r = E[(X - \mu)^r] &= E\left[\binom{r}{0}x^r - \binom{r}{1}x^{r-1}\mu + \binom{r}{2}x^{r-2}\mu^2 + \dots + (-1)^k \binom{r}{k}x^{r-k}\mu^k + \dots + (-1)^r \binom{r}{r}\mu^r\right] \\ &= \binom{r}{0}E(x^r) - \binom{r}{1}E(x^{r-1})\mu + \dots + (-1)^r \binom{r}{r}\mu^r\end{aligned}$$

If $a = \mu$ then

$$r = 1 \Rightarrow E[(X - \mu)] = 0$$

$$r = 2 \Rightarrow E[(X - \mu)^2] = \text{Var}(X)$$

Moments are called **central moments** according to the expected value, moments according to point 0 are called **ordinary moments**.

If $E[(X-\mu)^r] = \mu_r$ and $E[(X)^r] = m_r$;

Moments of the first three order

$$\mu_1 = E[X - E(X)] = E(X - m_1) = E(X) - m_1 = m_1 - m_1 = 0;$$

$$\mu_2 = E[X - E(X)]^2 = E[(X - m_1)^2] = E(X^2) - 2m_1E(X) + m_1^2 = m_2 - m_1^2;$$

$$\mu_3 = E[X - E(X)]^3 = E[(X - m_1)^3] = E(X^3) - 3m_1E(X^2) + 3m_1^2E(X) - m_1^3 = m_3 - 3m_1m_2 + 2m_1^3;$$

Example: Find the first 3 moments of the numbers 2,4,7,9,10,11 about 0.

Solution:

$$\begin{aligned}m_1 &= E[(X)^1] = 2\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 7\left(\frac{1}{6}\right) + 9\left(\frac{1}{6}\right) + 10\left(\frac{1}{6}\right) + 11\left(\frac{1}{6}\right) = 7,17 \\m_2 &= E[(X)^2] = 2^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 7^2\left(\frac{1}{6}\right) + 9^2\left(\frac{1}{6}\right) + 10^2\left(\frac{1}{6}\right) + 11^2\left(\frac{1}{6}\right) = 61,8 \\m_3 &= E[(X)^3] = 2^3\left(\frac{1}{6}\right) + 4^3\left(\frac{1}{6}\right) + 7^3\left(\frac{1}{6}\right) + 9^3\left(\frac{1}{6}\right) + 10^3\left(\frac{1}{6}\right) + 11^3\left(\frac{1}{6}\right) = 579,2\end{aligned}$$

Example Find the first 3 moments of the numbers 2,4,7,9,10,11 about mean.

Solution:

$$\begin{aligned}\mu_1 &= E[(X - \mu)] = 0 \\ \mu_2 &= E[(X - \mu)^2] = (2 - 7,2)^2 \frac{1}{6} + (4 - 7,2)^2 \frac{1}{6} + (7 - 7,2)^2 \frac{1}{6} \\ &\quad + (9 - 7,2)^2 \frac{1}{6} + (10 - 7,2)^2 \frac{1}{6} + (11 - 7,2)^2 \frac{1}{6} = 10,47 \\ \mu_3 &= E[(X - \mu)^3] = (2 - 7,2)^3 \frac{1}{6} + (4 - 7,2)^3 \frac{1}{6} + (7 - 7,2)^3 \frac{1}{6} \\ &\quad + (9 - 7,2)^3 \frac{1}{6} + (10 - 7,2)^3 \frac{1}{6} + (11 - 7,2)^3 \frac{1}{6} = -15,12\end{aligned}$$

MOMENT GENERATING FUNCTIONS

Definition: The moment-generating function of a random variable X is given by

$$M_X(t) = E[e^{tX}] = \begin{cases} \sum_{i=1}^k e^{x_i t} \cdot f(x_i) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{xt} \cdot f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Remark: By using Maclaurin's Series expansion, we have

$$e^{tx} = 1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^r x^r}{r!} + \dots$$

Then,

$$M_X(t) = E(e^{tx}) = 1 + tE(x) + \frac{t^2}{2!} E(x^2) + \frac{t^3}{3!} E(x^3) + \dots + \frac{t^r}{r!} E(x^r) + \dots$$

$$1 + tm_1 + \frac{t^2}{2!} m_2 + \frac{t^3}{3!} m_3 + \dots + \frac{t^r}{r!} m_r + \dots$$

Properties of Moment Generating Function

- 1) * If $t = 0$ is put in the 1st derivative of the moment generating function

$$M'_x(t = 0) = \left. \frac{dM_x(t)}{dt} \right|_{t=0} = E(X) = \mu = m_1$$

$E(X)$ is the first moment of X random variable.

- 1) * If $t = 0$ is put in the 2nd derivative of the moment generating function

$$M''_x(t = 0) = \left. \frac{d^2 M_x(t)}{dt^2} \right|_{t=0} = E(X^2)$$

$E(X^2)$ is the second moment of X random variable

- * If $t = 0$ is put in the n^{th} derivative of the moment generating function

$$M_x^{(n)}(t = 0) = \left. \frac{d^n M_x(t)}{dt^n} \right|_{t=0} = E(X^n)$$

$E(X^n)$ is the n^{th} moment of X random variable.

Note:
$$\text{Var}(X) = E(X^2) - [E(X)]^2 = M''_x(0) - [M'_x(0)]^2$$

2) If a and b are constants, then

i)
$$M_{X+a}(t) = E\left[e^{(X+a)t}\right] = e^{at} M_X(t)$$

ii)
$$M_{bX}(t) = E\left[e^{(bX)t}\right] = M_X(bt)$$

iii) If $Y=aX+b$, then

$$M_Y(t) = E(e^{tY}) = E[e^{t(aX+b)}] = e^{tb} E(e^{aXt}) = e^{tb} \cdot M_X(at)$$

iv)
$$M_{\frac{X+a}{b}}(t) = E\left[e^{\left(\frac{X+a}{b}\right)t}\right] = e^{\frac{a}{b}t} M_X\left(\frac{t}{b}\right)$$

3) The moment generating function is unique. This means that if two random variables have the same moment-generating function, they have the same probability distribution. If they have different moment generating functions, they can be said to have different distribution.

Example: Given that X has the probability function $f(x) = \frac{1}{8} \binom{3}{x}$ for $x=0,1,2,3$, find the moment generating function of the random variable X and use it to determine m_1 and m_2 .

Solution:

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \frac{1}{8} \cdot \sum_{x=0}^3 e^{tx} \binom{3}{x} \\ &= \frac{1}{8} (1 + 3e^t + 3e^{2t} + e^{3t}) \\ &= \frac{1}{8} (1 + e^t)^3 \end{aligned}$$

$$m_1 = M'_X(0) = \frac{3}{8} (1 + e^t)^2 e^t \Big|_{t=0} = \frac{3}{2}$$

$$m_2 = M''_X(0) = \frac{3}{4} (1 + e^t) e^{2t} + \frac{3}{8} (1 + e^t)^2 e^t \Big|_{t=0} = 3$$

Example: Find the moment-generating function of the random variable X whose probability function is

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

and use it to find the mean and the standard deviation.

Solution:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^{\infty} e^{tx} \frac{1}{2} e^{-x/2} dx = \frac{1}{2} \int_0^{\infty} e^{x(t-1/2)} dx$$

$$= \frac{1}{2} \frac{1}{(t-1/2)} e^{x(t-1/2)} \Big|_0^{\infty} = \frac{1}{2t-1} (0-1) = \frac{1}{1-2t} \quad \text{for } t < 1/2$$

WORKSHEET

1) Let A and B are any events. If

$$P(A) = 1/2, \quad P(B) = 1/3, \quad P(A \cap B) = 1/4$$

then compute the probabilities given below.

i) $P(A|B)$

ii) $P(B|A)$

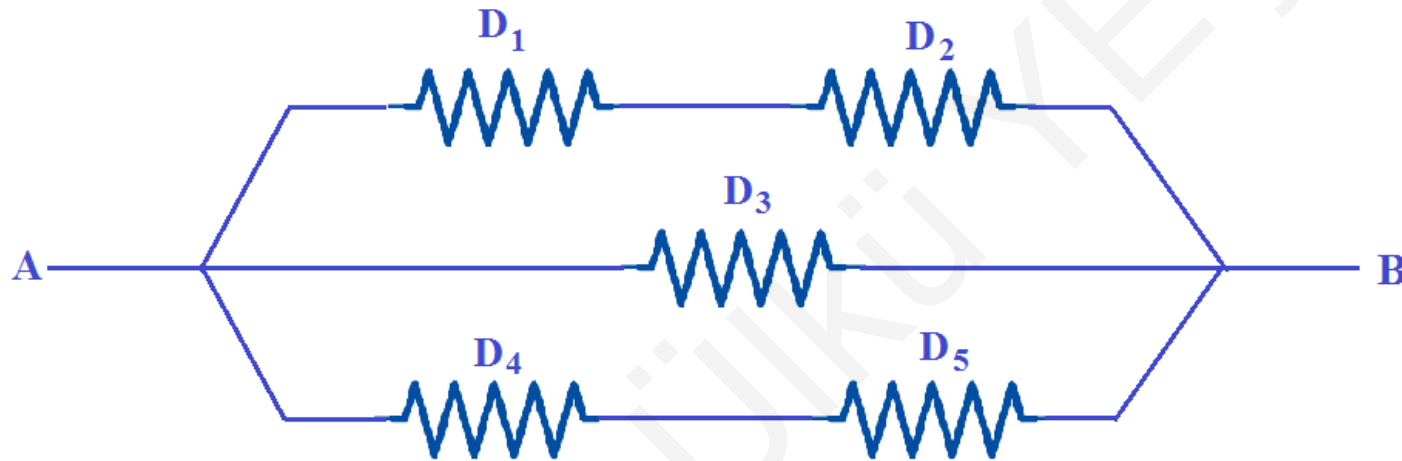
iii) $P(A \cup B)$

iv) $P(\bar{A}|\bar{B})$

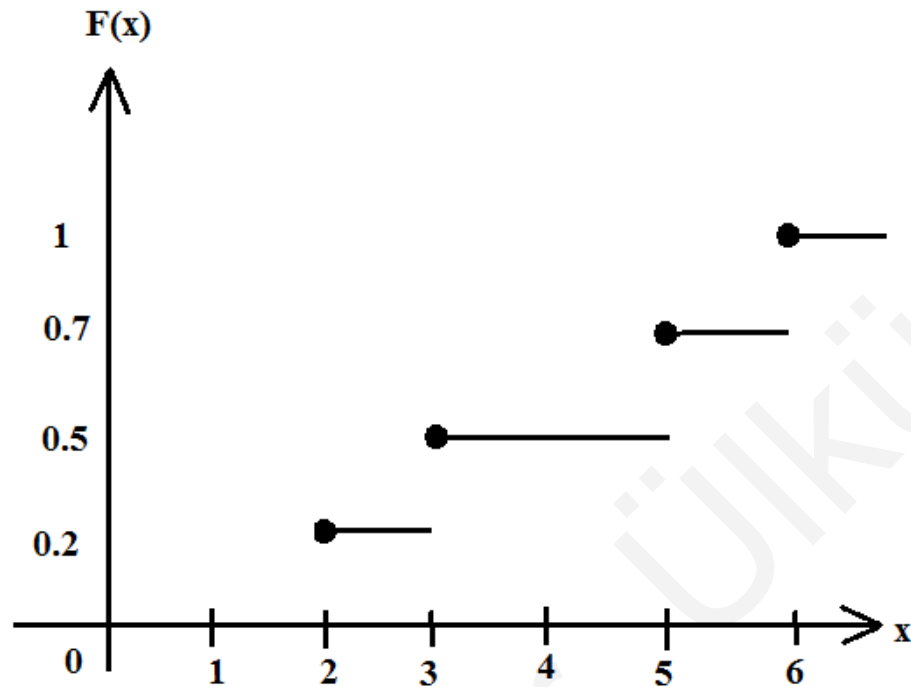
v) $P(\bar{B}|\bar{A})$

vi) $P(A \cap \bar{B})$

2) It is known that the current passes if the electrical switches are closed in an electrical circuit. In the following system, each of switches operates independently from each other. If the probability of switches being closed is 0.7, what is the probability that the current passes from A to B?



3) By using graph of cumulative distribution $F(x)$, find the probabilities;



i) $P(X < 2)$

ii) $P(X \leq 2)$

iii) $P(X > 3)$

iv) $P(X \geq 3)$

v) $P(X = 5)$

vi) $P(X = 6)$

4) If $f(x)$ is a probability function.

$$f(x) = \begin{cases} c(x+1), & 1 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

- a) Find the value of c .
- b) Determine the cumulative func.
- c) Find $P(1 \leq x \leq 2)$

5) If $F(x)$ is the cumulative distribution of X random variable; find the probabilities;

$$F(x) = \begin{cases} 0, & x < 0 \\ x/4, & 0 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

- i) $P(X \leq -1)$
- ii) $P(X \leq 1)$
- iii) $P(1 < X \leq 3)$
- iv) $P(X > 0.75)$

6) A loaded coin, the probability of showing tail is $\frac{2}{3}$ and showing head is $\frac{1}{3}$, is tossed three times. If random variable of X is the number of tails, what is the expected value of X ?

Ans: 2

7) A box contains 12 TV receiver and 2 of them have white caple. If 3 TV receiver are randomly selected in order to send to an hotel, what is the expectation of the number of chosen receivers with white cable?

Ans: 0.5

8) Given the probability density function

$$f(X) = \begin{cases} \frac{4}{\pi(1+x^2)}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

of the random variable X , find the expected value and the standard deviation of X .

$$(\text{Ans: } E(X) = \frac{\ln 4}{\pi} \cong 0.4413, \sigma \cong 0.2802)$$

9) A ball is randomly selected from a bag of N balls which are numerated from 1 to N . If X represents the numbers on the balls, find $E(X)$, $E(X^2)$ and σ_X^2 .

$$(\text{Ans: } E(X) = \frac{N+1}{2}, E(X^2) = \frac{(N+1)(2N+1)}{6}, \sigma_X^2 = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4})$$

10) 70% of the customers who come to a cake shop buy 2 bagels and leave the cake shop while 30% of them sit and drink coffee. 2/3 of those who sit and have coffee take 1 bagel and have breakfast as well as 1/3 of them take 2 bagels and have breakfast. 40% of the seats who have bought 1 bagel and 40% of the seats who have bought 2 bagel buy also 2 bagels when they leave. (60% of the seats do not buy bagels when they leave). Find the expected value of the number of bagels sold for each customer.

$$(\text{Ans: } E(X) = 2.04)$$